Choose by Your Data

01 / 03 | Common data models

Categories overlap: a single engine can support several of these.

01 | Relational

Connected Tables

MODEL
Rows, columns and relationships.

USE
Transactional applications.

CONSIDER
Schema, keys and joins.

EXAMPLE
SQL Server; PostgreSQL

Users

id (PK)	name
1	Anna
2	Luis

Orders

id	user_id (FK)
101	1
102	2

Orders.user_id references Users.id. Join matching keys.

02 | Key-Value

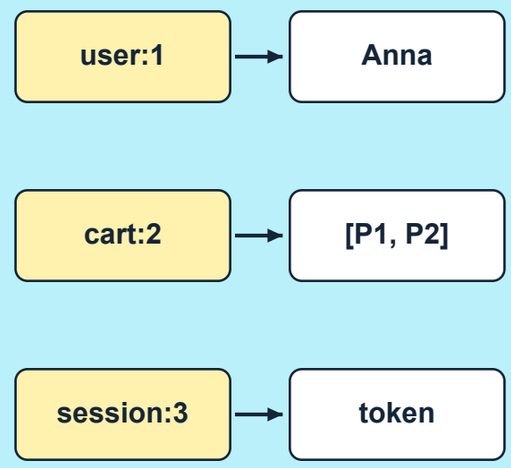
One Key, One Value

MODEL
Key / value pairs.

USE
Caching and sessions.

CONSIDER
Key-based access patterns.

EXAMPLE
Redis; Memcached



GET user:1 -> Anna
Keys identify stored values.

03 | Document

Nested Data

MODEL
Documents with nested fields.

USE
Catalogs and content.

CONSIDER
Embedding vs. references.

EXAMPLE
MongoDB; DynamoDB

```
{
  "id": "P1",
  "name": "Notebook",
  "attributes": {
    "color": "gray",
    "screen": 15.6
  },
  "tags": [
    "work", "portable"
  ]
}
```

An object can contain nested objects and arrays.

04 | Wide-Column

Sparse, Flexible Rows

MODEL
Row keys and column families.

USE
IoT and large datasets.

CONSIDER
Partition keys and queries.

EXAMPLE
HBase; Cassandra

Row key + column family

row	info:site	m:temp
S1	Room A	22.5
S2	Patio	--
S3	Office	20.1

info = information
m = metrics

Sparse rows need not populate every column. Not the same as columnar analytics storage.

05 | Property Graph

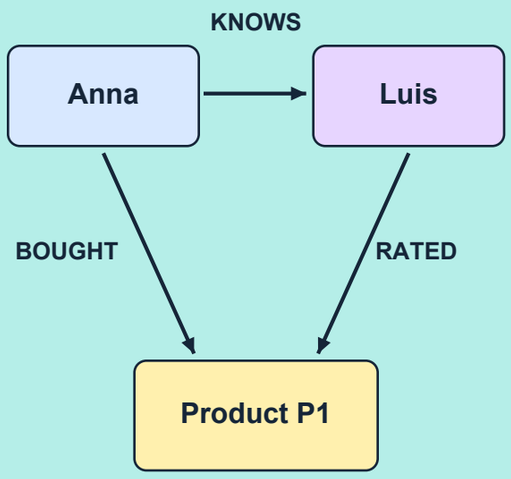
Relationships First

MODEL
Nodes, edges and properties.

USE
Networks and recommendations.

CONSIDER
Relationship traversals.

EXAMPLE
Neo4j



Nodes represent entities. Directed edges express specific relationships.

06 | RDF

Statements as Triples

MODEL
Subject, predicate, object.

USE
Knowledge graphs and semantics.

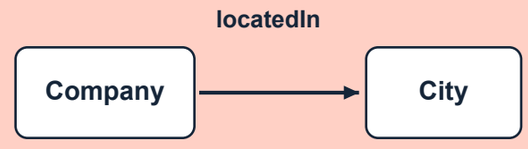
CONSIDER
Vocabularies and SPARQL.

EXAMPLE
Virtuoso; MarkLogic

Subject / predicate / object

ex:Anna
ex:worksAt ex:Company

ex:Company
ex:locatedIn ex:City



ex: is an example IRI prefix. RDF represents statements as triples.

Conceptual examples. Capabilities vary by engine.

Choose by Your Queries

02 / 03 | Specialized stores and capabilities

Categories overlap: a single engine can support several of these.

07 | Full-Text Search

Find Matching Terms

CAPABILITY
Analyzed terms and indexes.

USE
Text search and relevance.

CONSIDER
Analyzers and ranking.

EXAMPLE
Elasticsearch

```
D1: data graphs
D2: data tables
D3: graphs search
```

term	documents
data	D1, D2
graphs	D1, D3
search	D3
tables	D2

Query: graphs
Matching documents: D1, D3.
Each posting matches the text above.

08 | Vector Search

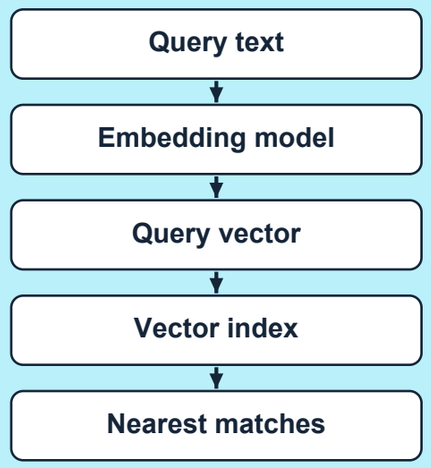
Search by Similarity

CAPABILITY
Vectors in a common space.

USE
Semantic search and recommendations.

CONSIDER
Embeddings and metric.

EXAMPLE
Pinecone; Chroma



Stored documents are embedded in a compatible vector space.

Illustrative 2D distances

A: 0.036 | B: 0.063

Nearest matches: A, B

09 | Spatial

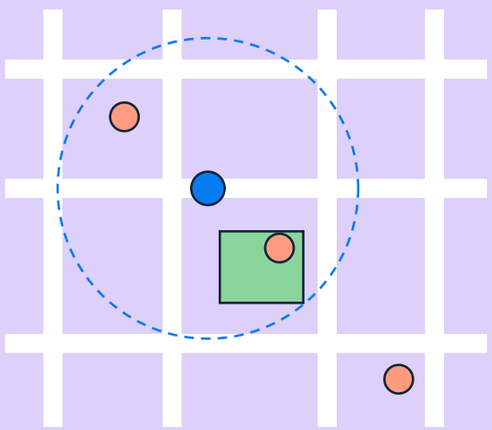
Query Locations

CAPABILITY
Points, lines and polygons.

USE
Maps and proximity.

CONSIDER
Coordinate systems and indexes.

EXAMPLE
PostGIS; SpatiaLite



Schematic planar map.
A radius query finds points within a specified distance.

10 | Time-Series

Time Is the Axis

MODEL
Timestamped observations.

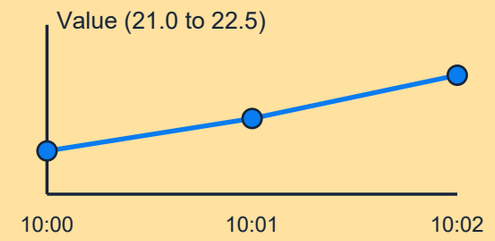
USE
Sensors and monitoring.

CONSIDER
Time windows and retention.

EXAMPLE
InfluxDB; Prometheus

Same sensor, ordered samples

time	value
10:00	21.4
10:01	21.7
10:02	22.1



11 | Native XML

Tags and Hierarchies

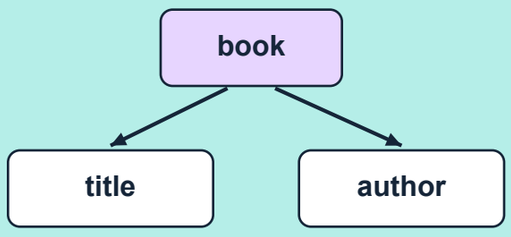
MODEL
XML documents and trees.

USE
Document integration.

CONSIDER
XPath and XQuery.

EXAMPLE
MarkLogic

```
<book>
  <title>Data</title>
  <author>Anna</author>
</book>
```



XPath: /book/author
Result: <author>Anna</author>

12 | Content Repository

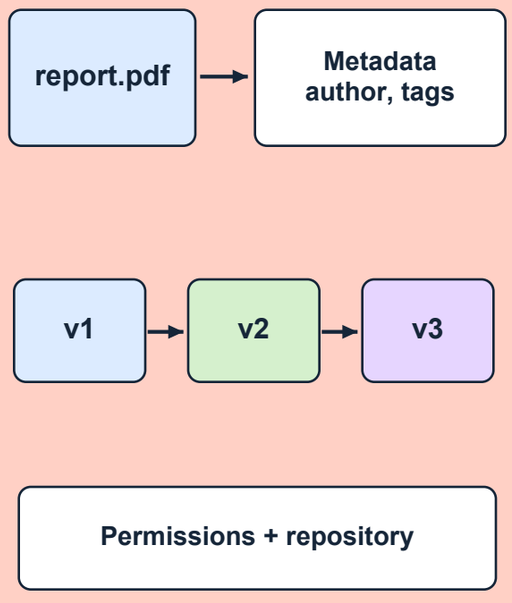
Version Your Assets

MODEL
Assets and metadata.

USE
Content management.

CONSIDER
Versions and permissions.

EXAMPLE
Apache Jackrabbit



Versioned assets include metadata and access rules.

Conceptual examples. Capabilities vary by engine.

Choose by Your Architecture

03 / 03 | Other models and architectural approaches

Categories overlap: a single engine can support several of these.

13 | Multivalue

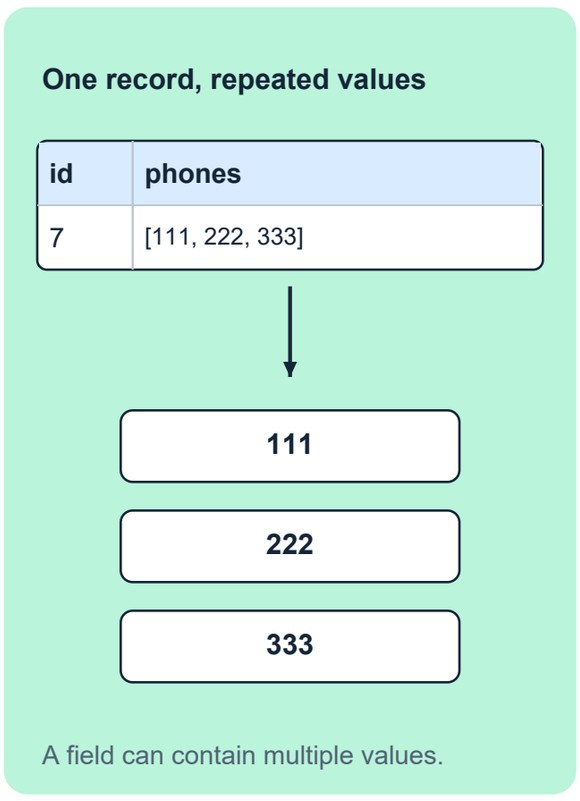
Repeated Field Values

MODEL
Fields with multiple values.

USE
ERP and inventory.

CONSIDER
Queries and relationships.

EXAMPLE
UniData; UniVerse



14 | Object-Oriented

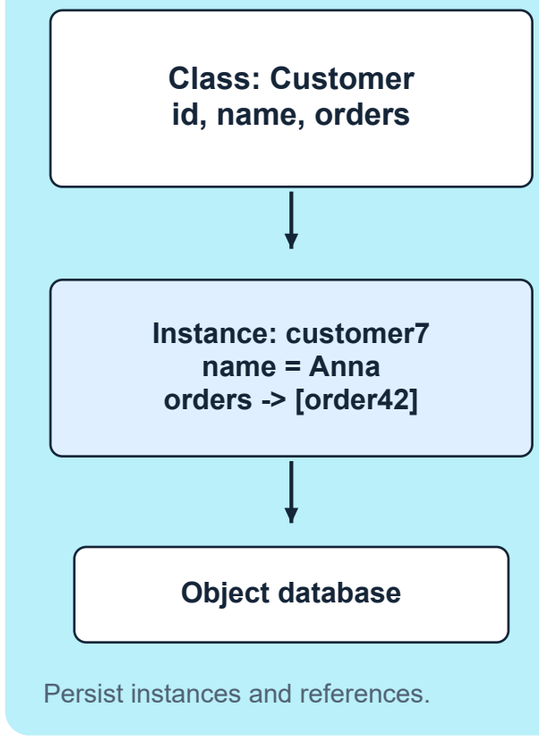
Persist Objects

MODEL
Objects and references.

USE
Complex object domains.

CONSIDER
Language and query support.

EXAMPLE
ObjectDB



15 | Navigational

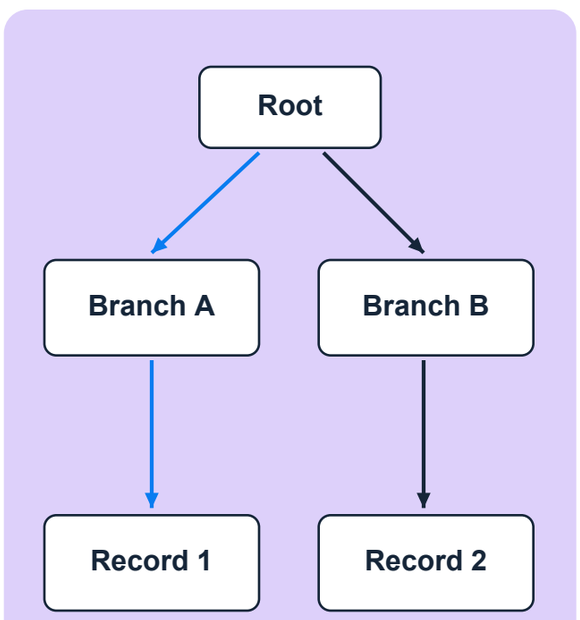
Follow the Links

MODEL
Records with predefined links.

USE
Hierarchical or network data.

CONSIDER
Access paths.

EXAMPLE
IBM IMS; IDMS



16 | Multimodel

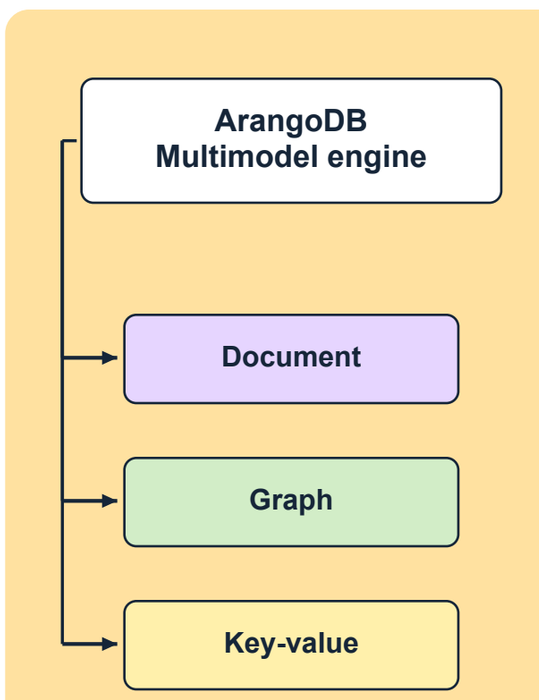
Combine Data Models

MODEL
Several models in one engine.

USE
Mixed data requirements.

CONSIDER
Models actually supported.

EXAMPLE
ArangoDB



17 | Event Sourcing

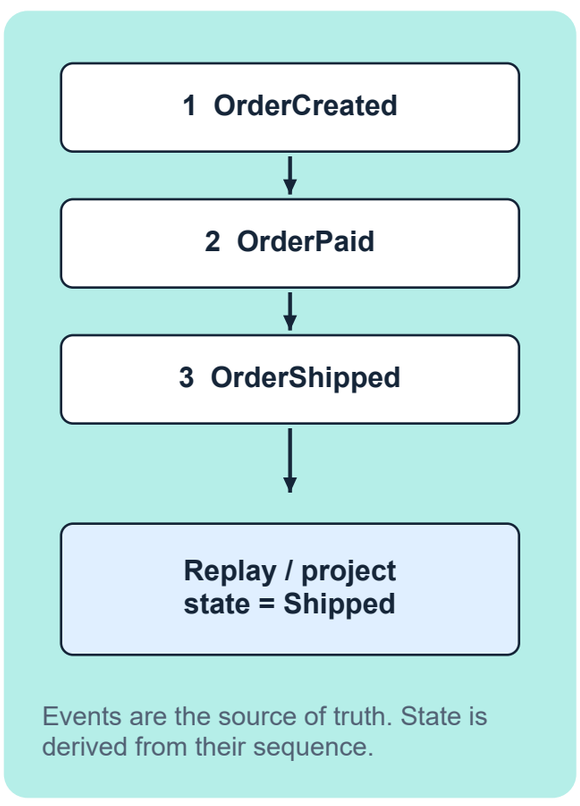
Rebuild the State

APPROACH
Ordered events as source of truth.

USE
History and state reconstruction.

CONSIDER
Ordering and projections.

EXAMPLE
KurrentDB (EventStoreDB)



18 | Blockchain

Link and Agree

APPROACH
Hash-linked blocks plus consensus.

USE
Shared distributed ledgers.

CONSIDER
Trust model and finality.

EXAMPLE
BigchainDB (source example)

